AgentISA: Formal Instruction Set Design for LLM Agents

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Abstract

Large Language Models run as autonomous agents, yet ad-hoc prompt engineering and per-provider function calling leave their behavior difficult to specify, analyze, and optimize systematically. We present AgentISA, an instruction set architecture making agent behavior a first-class compilable artifact: (1) a formal instruction set—compositional semantics, a type system, pre/post-conditions—checked offline by a shipped bounded verifier for pure instructions, bracketed by runtime pre-dispatch validation; (2) Agentic Calling, lifting provider-specific function calling into reusable, cacheable instructions; (3) instruction learning that mines recurring trace patterns, promoting them as composite-replay instructions; (4) a compiler and runtime applying classical optimization—fusion, dead-code elimination, pipeline scheduling, dynamic hot-path specialization—to agent instruction streams.

A reproducible microbenchmark drives the real implementation against a mock LLM backend: result caching reduces simulated API calls by 59.5–79.6% on synthetic Zipf workloads (2,000 tasks, 5 seeds); static compilation cuts the effective instruction count by 34.2% (35.4% at length 1,000), decomposed by a same-seed ablation into fusion (21.6%) and dead-code elimination (24.5%) with an 11.9-point overlap; dynamic hot-path optimization shrinks repeatedly executed traces by 65.8%. Real-gateway experiments extend validation across three production models—the cache reproduces computed hit/miss statistics exactly on both models tested with measured time and token savings, and the compiled, hot-path-adopted stream issues zero new gateway calls with byte-identical artifacts, live-model classifications included in one rerun (cache-mediated)—and a four-task system-level characterization (12.5–25.0% per-task compile shrink) calibrates the synthetic headline; a 100-trace census (792 steps, 12 instruction signatures) confirms Zipf-like reuse outside the hand-built benchmarks ($s = 0.593$, $R^2 = 0.971$; 98.5% of instances recur corpus-wide, per-workflow duplication averaging 18.3%); 43 multi-agent tests provide functional validation. We position these results as controlled-setting evidence, not production end-to-end measurements. Result caching, the benchmark-firing static passes, and threshold-triggered trace re-optimization ship; capability metric extraction, parameterized replay, opt-in tree-pattern selection, EMA hot-path detection, strict security, and cache namespacing now also ship, each exercised by a controlled mechanism experiment on the real code path (Section 5.11). Cycle-accurate pipeline scheduling, block-granular trace formation, and symbolic-equivalence checking remain specified—no claim rests on them.

CCS Concepts

• **Software and its engineering** → >**Compilers**; • **Theory of computation** → >**Program reasoning**; • **Computing methodologies** → >**Artificial intelligence**; • **Computer systems organization** → >**Architectures**.

Keywords

Instruction Set Architecture; Large Language Models; Compiler Optimization; Multi-agent Systems; Formal Methods; Result Caching

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1 Introduction

Recent advances in Large Language Models (LLMs) have produced systems with strong understanding, generation, and—through instruction tuning and human feedback—instruction-following capabilities [1, 2, 3, 4, 5, 6], and a rapidly growing body of work couples them with tools, memory, and control loops to build autonomous agents [9, 10, 11, 12]. Yet the engineering substrate beneath these agents remains ad hoc: behaviors are specified as prompts and provider-specific function-calling schemas—difficult to specify precisely, analyze formally, or optimize systematically.

1.1 Background and Motivation

Function calling—selecting an API from a declared tool set and emitting structured arguments—has become the dominant interface between LLMs and external systems [13, 14, 15], consolidated by standardization efforts such as the Model Context Protocol [18] and measured by harnesses such as the Berkeley Function Calling Leaderboard [17]. However, the unit of interaction remains the individual API call: each call is dispatched, priced, and forgotten independently, even when an agent repeatedly executes structurally identical work.

This per-call statelessness contrasts with how computing systems historically handle repetitive execution: classical compilers and hardware translate recurring patterns into cheaper forms—instruction fusion, dead-code elimination, caching, trace-based dynamic optimization [33, 34, 45]. Our motivating hypothesis is that these techniques apply to LLM agent workloads: if agent behavior is expressed as an explicit instruction stream rather than opaque prompts, recurring patterns become visible to a compiler

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and optimizable with well-understood, semantics-preserving transformations.

1.2 Challenges in Current LLM-based Agent Systems

Five structural challenges motivate AgentISA:

1. **Per-call statelessness of tool interfaces.** Mainstream function-calling protocols treat each call as an isolated event [13, 14, 18]: the provider has no notion of a *program* being executed, repeated identical work incurs repeated cost, and caching, where present, is a bolt-on afterthought rather than a property of the execution model.

2. **Experience that does not accumulate as compiler-visible instruction structure.** Paradigms such as Reflexion [10] and Voyager [12] accumulate experience as verbal reflections or inference-time skill libraries—not reified as reusable, statically optimizable instructions a compiler can fuse, schedule, or eliminate.

3. **Absence of a shared behavioral contract.** Each multi-agent framework defines its own abstractions for roles, messages, and workflows [19, 20, 21, 22, 23, 24]; no instruction-level contract spans frameworks, fragmenting tooling and cross-framework analysis.

4. **Cost pressure at the wrong layer.** FrugalGPT [28] shows LLM inference cost is a first-order concern, and serving systems attack it below the API boundary [38, 39, 40, 41]; what is missing is the complementary, above-the-API layer that reduces the *number and size* of calls an agent issues in the first place.

5. **Integration complexity across providers.** Portability is handled at the library level [26, 27]: logic written against one provider's function-calling surface does not transfer cleanly to another, and capability differences are hard to reason about systematically [17].

1.3 Research Objectives and Contributions

This paper introduces AgentISA, which addresses these challenges by making agent behavior a first-class compilable artifact. Our contributions are:

1. **A formal instruction set** for LLM agents with compositional semantics, a type system, and pre/post-conditions on every instruction (Section 3.1). Two deliberately modest sanity results serve as design rationale: Turing completeness (Theorem 3.1—negative certification that the instruction set does not constrain expressiveness) and type safety for the pure core (Theorem 3.2), fixing the well-formedness obligation the offline bounded checker discharges for pure instructions (Section 3.3.3) and the runtime brackets with pre-dispatch validation.

2. **Agentic Calling**, a mechanism lifting provider-specific function calling into reusable, cacheable, semantically specified agent instructions, with a model-adapter layer for cross-provider portability (Section 3.2).

3. **Trace-based instruction learning**, which mines recurring instruction patterns from interaction traces and promotes validated recurring patterns into candidate instructions whose `execute` bodies replay the mined constituent subsequence by sequential composition over a shared context (Section 3.3; replay reproduces the recorded subsequence's outputs on the recorded inputs and, via shipped schema-checked binding, on unseen parameter values—constructor-baked parameters excepted, as the section states; the generalization is quantified: 1,050/1,050 unseen-parameter replays match the oracle exactly, Section 5.11).

4. **A compiler and runtime** implementing fusion, dead-code elimination, parallel-grouped scheduling, and dynamic hot-path specialization for agent instruction streams (Sections 3.4 and 4), together with an opt-in tree-pattern instruction selector (shipped, Section 4.1.1; benchmarked in Section 5.11: covering cost at or below the default heuristic's on 20/20 random IR trees, mean ratio 0.361, the gap synthetic-pricing-dependent) and cycle-accurate pipeline scheduling (specified, Section 4.1.2), with semantic-preservation arguments for the implemented rewrites (Theorems 3.4–3.6, proofs in Appendix C).

5. **An empirical evaluation** of the implemented system, with full methodology disclosure. On synthetic Zipf workloads (5 random seeds each), caching reduces simulated API calls by 59.5–79.6% depending on skew, static compilation reduces the effective instruction count by 34.2% ± 1.8% (length 200), and dynamic optimization shrinks repeatedly executed traces by 65.8% ± 4.6%; real-gateway experiments (Sections 5.7–5.7.1) validate the cache and the compiled pipeline against a real LLM gateway, a four-task system-level characterization (Section 5.8) composes the mechanisms on realistic (non-synthetic) streams, a 100-trace census (Section 5.10) tests whether Zipf-like reuse and the compiler's fusibility assumptions hold outside the hand-built benchmarks, and six controlled mechanism experiments (Section 5.11) exercise the shipped opt-in components.

6. **Multi-agent coordination infrastructure** with capability-based agent selection, execution planning, fallback strategies, and a security monitor enforcing authentication, authorization, resource-access, and data-sensitivity policies (Sections 3.5 and 5.6), validated by 43 passing functional tests; an opt-in fail-closed security mode now also ships, and its paired posture matrix (1,000 evaluation tasks per mode) shows it denying the 400 policy-gap tasks the fail-open default admits with zero false denials on well-specified tasks (Section 5.11).

Our empirical claims are scoped accordingly: Sections 5.2–5.5 exercise the real `agentflow.core.isa` implementation against a mock LLM backend on synthetic workloads; Sections 5.7–5.8 extend the

validation at reduced scale on realistic streams; Section 5.10's census covers generator-produced traces and Section 5.11's mechanism experiments run on declaration-injection pools and synthetic trees—still controlled settings, none production end-to-end measurements (threats to validity in Section 5.9).

2 Related Work

2.1 Foundation LLMs and Function Calling

Modern LLMs are autoregressive transformers trained at large scale [1, 2, 3, 4], with instruction tuning and reinforcement learning from human feedback making them reliable instruction followers [5, 6]. Function calling constrains their output distribution to a structured output space, $F(x) = \text{argmax}_{f \in \mathcal{F}} P(f|x, \mathcal{F})$, where $\mathcal{F}$ is the set of declared functions and $F(x)$ the selected call. Frontier and open models alike expose this capability with provider-specific schemas [2, 3, 4]; AgentISA does not modify this layer but treats any such interface as a *backend* targeted by a common instruction set, as a compiler targets multiple processor backends [33].

The cost structure of this layer motivates our work. If each call carries latency τ_i plus a context-switching overhead ω_i (prompt reconstruction, serialization, cache misses; the symbol δ is reused in Appendix B for per-instruction transition functions—the two are unrelated), then n similar calls cost $C(n) = \sum_{i=1}^{n}(\tau_i + \omega_i)$.

Definition 2.1 (Function Call Overhead): The overhead of executing n function calls is $C(n)$, the sum of per-call latencies and per-call context reconstruction costs. When repeated calls carry identical semantic inputs and satisfy the cache-validity conditions of Section 3.2, the repeated context-construction cost ω_i is avoidable—which is precisely the cost AgentISA's caching and fusion are designed to remove. Structural similarity alone does not make a re-invocation waste: if an input's dependencies have changed or the call probes external state, re-execution has semantic meaning; the cache-validity conditions, not structural identity, license reuse.

2.2 Reasoning and Agent Paradigms

A second line enhances reasoning at prompt time: chain-of-thought [7] and zero-shot reasoning [8]; ReAct's interleaved reasoning and action [9]; Reflexion's verbal self-critique [10]; Tree of Thoughts' deliberative search [11]; Voyager's open-ended skill acquisition [12]. These paradigms define *what* an agent should do at the cognitive level; AgentISA defines *how* the resulting behaviors are represented, compiled, and executed as instructions—a ReAct loop becomes an instruction sequence whose tool invocations are fused, cacheable agentic calls.

2.3 Tool Calling and Interface Standardization

Toolformer trains models to interleave tool calls with text [13]; Gorilla connects models to large API corpora [14]; ToolLLM scales tool coverage to 16,000+ real-world APIs [15]; HuggingGPT orchestrates model hubs [16]. On the standardization side, the Model Context Protocol specifies how tools and context are exposed [18], and the Berkeley Function Calling Leaderboard evaluates function-calling accuracy cross-model [17]. These efforts standardize the *interface*; AgentISA standardizes the *execution model above it*: an AgentISA program compiled for two providers differs only in the final translation step (Section 3.2.3), and interface-level standards alone cannot fuse or eliminate calls.

2.4 Multi-Agent Frameworks

AutoGen frames agent applications as multi-agent conversations [19]; MetaGPT assigns standardized roles and workflows [20]; AgentVerse facilitates multi-agent collaboration and studies emergent behaviors [21]; ChatDev organizes agents as a software-development organization [22]; CAMEL studies communicative agent societies [23]; AutoGPT popularized autonomous goal-driven agents [24]. AgentISA contributes a different artifact to this space: not a framework or orchestration philosophy but an instruction-level substrate—capability profiles, coordination instructions with barrier semantics, and a security monitor providing capability-based agent selection, per-stage fallback planning, and policy enforcement—that any such framework could target (Section 3.5, Section 5.6).

2.5 Prompt Compilation and Cost Optimization

Closest to our compilation perspective is DSPy, which compiles declarative prompt programs into optimized pipelines via prompt search and fine-tuning [25]; LangChain [26] and LlamaIndex [27] provide library-level chaining and indexing abstractions; FrugalGPT reduces cost through model cascades [28]. AgentISA differs in the level at which optimization operates—DSPy optimizes *prompts*, FrugalGPT *model selection*, AgentISA *instruction streams*: fusion, dead-code elimination, parallel grouping (the shipped form of pipeline scheduling), result caching, and dynamic hot-path specialization—implemented and measured (Sections 3.4, 4, and 5)—plus tree-pattern instruction selection (shipped as an opt-in strategy, Section 4.1.1) and cycle-accurate pipeline scheduling (specified, Section 4.1.2). These classical compiler transformations [33] compose with—rather than replace—prompt-level and model-level optimization.

2.6 Benchmarks for Agent Systems

AgentBench evaluates LLMs as agents across diverse environments [29]; GAIA tests general assistant abilities [30]; PlanBench probes planning and reasoning about change [31]; SWE-bench measures resolution of real-world GitHub issues [32]. These benchmarks evaluate *end-to-end capability*; AgentISA's evaluation measures *subsystem behavior under controlled conditions*—call reduction, instruction-count reduction, trace shrinkage (Section 5). The two are complementary: instruction-level metrics can attribute end-to-end gains to specific mechanisms when such gains are measured, and are reproducible without live model access.

2.7 Classical Compiler and Architecture Techniques

AgentISA's optimization machinery is deliberately classical: tree-pattern instruction selection with dynamic programming,

peephole optimization, and data-flow-based dead-code elimination [33]; pipelining, hazard detection, and resource-constrained scheduling [34]; the open, well-specified instruction-set discipline of RISC-V [35]; type safety via progress-and-preservation [36]; formally verified compiler correctness in the CompCert tradition [37]. Our contribution is not a new technique but the demonstration that this machinery transfers to LLM agent workloads, with the semantic-preservation arguments adapted to instructions with pre/post-conditions (Appendix C).

2.8 LLM Serving Systems

Below the API boundary, the serving community has made inference dramatically cheaper: PagedAttention [38]; Orca's continuous batching [39]; FlashAttention's IO-aware attention [40]; Mooncake's KVCache-centric disaggregation [41]; inference-cost energy analyses [42]. This line and AgentISA are orthogonal layers of the same stack: serving systems make each call cheaper; AgentISA reduces how many calls an agent makes and how much redundant work each carries—the two compound multiplicatively.

2.9 Research Gaps and Our Contributions

In summary, the literature leaves four gaps that AgentISA addresses:

1. **No instruction-level contract for agent behavior.** Interfaces are standardized [13, 14, 18] and frameworks numerous [19, 20, 21, 22, 23, 24], but no shared instruction set spans providers and frameworks.

2. **No compiler for agent behavior.** Prompt-level compilation exists [25] and model-level cost reduction exists [28], and *caching itself is well-trodden ground*—at the response level (GPTCache [46]), the KV-prefix level (RadixAttention/SGLang [47]), and the provider prompt-caching level. None of these layers expose an *instruction-level* contract: caching there optimizes model serving—invisible to the agent program's structure, without effect on it. Instruction-level fusion, elimination, and scheduling—with cacheability a declared property of the instruction set—have not, to our knowledge, been demonstrated.

3. **No semantics-preserving account of agent optimization.** Serving work verifies kernels and schedulers, not agent-level transformations; we provide progress-and-preservation-style arguments for ours (Theorems 3.2, 3.4, 3.5, 3.6).

4. **No reproducible instruction-level evaluation.** Existing benchmarks measure end-to-end capability [29, 30, 31, 32]; we contribute a seed-controlled evaluation: a microbenchmark suite with fully disclosed synthetic workloads and mock-backend instrumentation, extended by real-gateway and system-level validations (Section 5).

3 AgentISA Design

Building on Sections 1 and 2, we present the AgentISA design: the core architecture (Section 3.1), the Agentic Calling framework (Section 3.2), trace-based instruction learning (Section 3.3), instruction organization and optimization (Section 3.4), and the capability-assessment layer (Section 3.5).

3.1 Core Architecture

3.1.1 Instruction Set Specification. The AgentISA instruction set is formally defined as a tuple:

Definition 3.1 (AgentISA Core): AgentISA $= (I, T, \Sigma, \Delta, \Phi)$, where I is the set of instructions, T the type system, Σ the state space, Δ the transition function set, and Φ the verification function set.

Following the conventions of classical instruction set design [33, 34, 35], we define the basic instruction format:

```
INSTRUCTION = {
    opcode: OPCODE,
    args: ARGS[],
    precondition: PRED,
    postcondition: PRED,
    context: CONTEXT
}
```

The instruction set is categorized into four fundamental classes: **control flow** ($\mathcal{C} = \{seq, branch, loop, call\}$), **state management** ($\mathcal{S} = \{load, store, update, cache\}$), **LLM interaction** ($\mathcal{L} = \{query, response, context\}$), and **resource management** ($\mathcal{R} = \{allocate, free, monitor\}$).

Two sanity results anchor the definition; neither introduces new machinery [33, 36]—Theorem 3.1 certifies that the instruction set does not constrain what agent programs can express, and Theorem 3.2 fixes the well-formedness obligation discharged by the offline bounded checker for pure instructions (Section 3.3.3) and bracketed at runtime by pre-dispatch context validation (Section 3.2.3).

Theorem 3.1 (Turing Completeness): The AgentISA instruction set is Turing complete.

Proof sketch: a translation from WHILE programs—known Turing equivalent—to instruction sequences, by simulation: state management instructions encode an unbounded store, control-flow instructions encode composition, branching, and iteration. The full construction is in Appendix A.

3.1.2 Type System and Semantics. The type system T extends the simply-typed discipline of Pierce [36] with dependent-type constructions (Π, Σ, μ):

Definition 3.2 (AgentISA Type System):

$$T ::= \text{Base} \mid \Pi(x : T_1).T_2 \mid \Sigma(x : T_1).T_2 \mid \mu X.T$$

where Base types include {Bool, Int, String, Float}, Π/Σ represent dependent function/pair types, and μ recursive types. The semantic rules follow standard operational semantics (a context rule and β-reduction).

The pure core carries a standard type-safety argument—progress and preservation, in the style of Pierce [36]—discharging the well-formedness role fixed in Section 3.1 and extended to the shipped instruction set by the bounded checker of Section 3.3.3:

Theorem 3.2 (Type Safety for the Pure Core): Well-typed programs in the pure core calculus of Appendix B—the λ-fragment

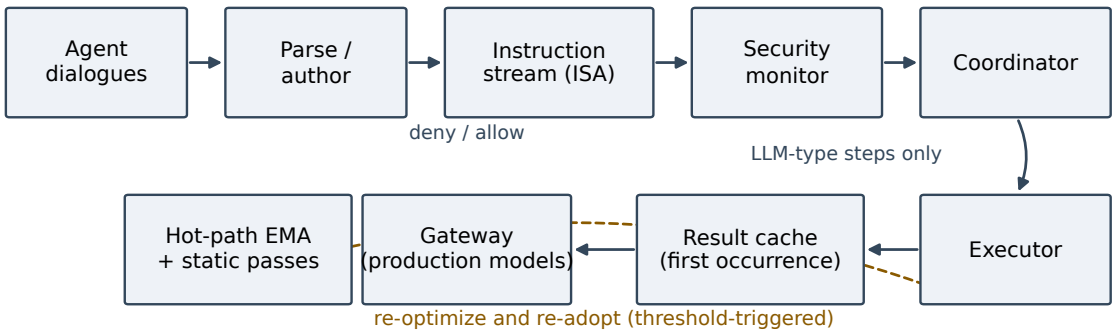


Figure 1: AgentISA runtime architecture. Authored streams pass a security gate and a coordinator that routes LLM-type steps through a result cache (first occurrence only) against the production gateway; a hot-path detector and the static optimizer re-optimize and re-adopt the stream (threshold-triggered). Sections 5.7-5.7.1 measure each property on the real code path.

(base types, dependent Π/Σ types, recursion) plus the composition operators applied to instructions carrying typed pre/post-conditions—cannot "go wrong", under the per-instruction validity obligation of Definition 3.5 (stated in Section 3.3.3 below; pre/post-conditions), with transition functions assumed total on precondition-satisfying states (Appendix B.7). Effectful instruction classes are embedded via their typed pre/post-conditions and covered to the extent their preconditions are checkable.

Proof sketch: By the standard two-lemma technique—progress ($\forall e, T.\ \vdash e : T \implies \text{value}(e) \vee \exists e'.\ e \to e'$) and preservation ($\forall e, e', T.\ \vdash e : T \wedge e \to e' \implies \vdash e' : T$)—as presented in Pierce [36], both proved by induction over Appendix B's merged judgment system, with a joint argument for the composition cases: progress from the Definition 3.5 obligation plus totality, preservation from postcondition typing, composition via the chained invariant $post(i_1) \Rightarrow pre(i_2)$ (Appendix B).

3.1.3 LLM Integration Interface. The LLM integration interface is a protocol playing the same role for agent backends that the Model Context Protocol plays for tool exposure [18]: $\mathcal{P}_{LLM} = (Q, R, C, \psi)$, where Q is the query space (JSON Schema), R the response space (JSON Schema), C the context-management surface (today the context-window size), and ψ the state mapping Map<AgentState, LLMContext>—corresponding to the interface conventions of modern function-calling providers [3, 18].

3.2 Agentic Calling Framework

3.2.1 Function Calling to Agentic Calling Transformation. The transformation from function calling to Agentic Calling is formalized as **Definition 3.3** (Agentic Transformation): $\mathcal{T} : F \to A$, where F is the function call space and A the agent instruction space.

$\mathcal{T}$ is required to preserve call semantics; this holds by construction for the three-step process below—the synthesized instruction records the function signature verbatim, embeds the context verbatim (Theorem 3.3), and delegates execution to the original call—so we state preservation as a property of the construction rather than a separately proved theorem: $\forall f \in F : \text{sem}(\mathcal{T}(f)) \equiv \text{sem}(f)$.

The three steps—function signature analysis, context embedding, instruction synthesis—mirror the signature-to-call conventions of function-calling interfaces [13, 14].

3.2.2 Context Preservation and State Management. State management in AgentISA follows a formal state machine model, $M = (S, s_0, \delta, F)$: states S with initial state s_0, transition function δ, and final states F.

Extraction-time context preservation is guaranteed by construction (Theorem 3.3); at runtime it is only monitored:

Theorem 3.3 (Context Preservation): Let ϕ be the context interpretation function (defined compositionally over context *contents*—the keys and values the context carries), and t the extraction transformation that embeds a context c verbatim into an instruction's `context` field. Then, for every $c \in C$: $\phi(t(c)) = \phi(c)$.

Proof: By the definition of t, the extracted instruction carries a verbatim copy of c; since ϕ is defined over context contents and the copy is content-identical, $\phi(t(c)) = \phi(c)$ holds by construction—a *design invariant of the extraction pipeline* (the implementation copies rather than re-encodes). At runtime the shipped checks are constraint-based and bracket the transition—`AgenticTransformer.validate_context` checks the registered context constraints pre-dispatch, `verify_context_preservation(before, after)` re-checks them pairwise across the executed transition (mechanics in Section 3.2.3)—and perform no structural validation of the carried context. Extraction-time preservation is discharged by the copy construction; preservation across an executed transition is *monitored, not guaranteed*, by the runtime checks. ■

3.2.3 Cross-model Compatibility. Cross-model *protocol* compatibility—not behavioral equivalence of models—is achieved through an abstract interface layer (provider-independent instruction representation with provider-specific adapters):

```
Interface ModelAdapter {
    translate: AgentInstruction → ModelSpecificCall      // shipped:
    ↪ translate_instruction
    normalize: ModelSpecificResponse → AgentResponse    // shipped:
    ↪ normalize_response
```

```
    validateContext: Context → Boolean                // shipped:
    ↪ AgenticTransformer.validate_context
}
```

All three operations ship: `translate` is `ModelAdapter.translate_instruction` (with registered type transformations); `normalize` is `ModelAdapter.normalize_response`, folding the two dominant wire shapes—Anthropic (typed `content` blocks, `usage.input_tokens/output_tokens`) and OpenAI (`choices[0].message`, `usage.prompt_tokens/completion_tokens`)—into a canonical `AgentResponse` envelope (content, tool calls, usage, model, raw), so downstream code never branches on provider syntax; and `validateContext` is `AgenticTransformer.validate_context`, the pre-execution complement of the shipped post-hoc `verify_context_preservation(before, after)`—it evaluates the registered context constraints against a single state before dispatch, applying two-argument predicates reflexively (v, v) (single-argument ones directly) and leaving constraints over absent keys vacuously satisfied, so key *deletion* goes unreported—a blind spot analogous to non-reflexive constraints (e.g. strict growth), which are not detectable pre-execution.

Motivated by cross-model function-calling studies [17] and open protocol efforts [18]—providers differ in surface syntax but can agree on *protocol shape*—the requirement is a **protocol-level contract on the adapter layer**: each adapter m's request and response normalizers, $\text{norm}_{in}(m)$ and $\text{norm}_{out}(m)$, are total on the supported request and response classes, mapping every supported raw request and response into the unified schemas, with declared types preserved and errors classified into a common taxonomy. This is deliberately weaker than semantic-output equivalence—$m_1 = m_2$ would already demand determinism, which no LLM provides—and cross-model *behavioral* comparison is an empirical question at benchmark level (Section 3.5, [17, 29]).

3.3 Trace-Based Instruction Learning

3.3.1 Dialogue Pattern Recognition. Pattern recognition in AgentISA is frequency-based and deterministic (action-signature grouping) rather than grammatical; it is implemented in `agentflow.core.isa.pattern_mining` and `agentflow.core.isa.instruction_learning`. To prevent over-reading: the subsystem is a **discovery-and-replay pipeline, not an autonomous learning loop**. Its five stages are (i) trace-derived pattern mining (this section), (ii) dialogue-derived template extraction (§3.3.2), (iii) candidate validation (support and success-rate thresholds), (iv) *gate-controlled* promotion (a shipped proposal engine, `agentflow.core.isa.auto_promotion`, edits the optimizer's rule set only through an approval gate; an opt-in `auto_approve` removes the human step), and (v) parameterized composite replay (re-running the recorded subsequence with caller-supplied, schema-checked parameter values). Stages (i)–(v) ship; the stage-(iv) gate defaults to approval.

Definition 3.4 (Instruction Sequence Pattern): A pattern p over the instruction alphabet I is a non-empty contiguous sequence $p = (i_1, \ldots, i_k) \in I^k$ together with two frequency statistics:

- **Support**: $sup(p) = P(p \text{ occurs as a contiguous subsequence in a randomly drawn trace})$
- **Confidence**: $conf(p \Rightarrow j) = P(i_{t+1} = j \mid (i_{t-k+1}, \ldots, i_t) = p)$

A pattern is *frequent* if $sup(p) \geq \sigma_{min}$ and *predictive* if $\max_j conf(p \Rightarrow j) \geq \gamma_{min}$, for thresholds $\sigma_{min}, \gamma_{min} \in (0, 1]$.

Pattern recognition proceeds in three stages:

1. **Frequent-sequence mining.** Candidate subsequences are enumerated from execution traces with occurrence counts; successor statistics are accumulated alongside, and sequences below σ_{min} are pruned together with their statistics—so confidence is estimated only over frequent sequences. The output is the set of frequent, predictive patterns.

2. **Behavior grouping.** The shipped `PatternRecognizer` groups dialogues deterministically by *action signature*—the sorted set of system-turn contents—and promotes each sufficiently supported bucket into a pattern. An earlier prototype (TF-IDF embeddings, density-based clustering, k-means) was abandoned: non-deterministic across library versions and degenerate on the short traces agents provide (a k-means view over instruction-trace features remains in `pattern_mining`, fixed `random_state`). Promotions split by source: trace-mined promotions execute by **composite replay**—the `execute` body re-runs the recorded constituent sequence, with its dataflow edges, in mined order against a shared context, merging outputs and returning the keys its `produces_X` postconditions declare—while dialogue-mined templates run an attached `executor` if one exists (dict results checked against the declared `produces_X` postconditions) and otherwise raise `NotImplementedError` rather than silently no-opping. Promotion is deterministic (pattern identity derives from the mined instruction-id sequence), and replay is plain sequential composition—nothing smarter—so a learned instruction reproduces its recorded subsequence's outputs on the recorded inputs; on unseen parameter values the binding above re-runs the recorded operator *instances*, so parameters baked into an operator's constructed state cannot be rebound, and the schema is observational—mixed or unseen types fall back to `object`. The mining-to-optimization path now exists: an approved proposal registers the fused instruction and installs a name-level peephole rule into the optimizer's rule set, gated by default on a developer's approval.

3. **Pattern validation.** Each candidate is checked for significance—support above σ_{min} and, for dialogue-derived patterns, an acceptable success rate—before promotion to a candidate instruction.

The design is deliberately statistical—agent traces are short and noisy, and frequency statistics degrade gracefully where grammar induction would overfit—and it composes with the compiler: a frequent, verified pattern is exactly the promotion candidate of Section 3.4.4.

3.3.2 Instruction Extraction Pipeline. The shipped extraction process instantiates the Section 3.3.1 pipeline: frequent-sequence mining, deterministic action-signature grouping, and threshold-gated promotion with the standard optimization passes applied to any resulting stream.

3.3.3 Instruction Optimization and Verification. Optimization is coupled to verification: where verified-compilation efforts such as [37] verify the transformations of a whole compiler, AgentISA's verification targets a different granularity—the validity of individual instructions and the obligations of individual rewrites:

Definition 3.5 (Instruction Verification): An instruction i is valid if $\forall s \in S : \text{pre}(i, s) \implies \text{post}(i, \text{exec}(i, s))$.

The shipped checker (`agentflow.core.isa.bounded_checker`) enumerates the candidate input states derived from the declared parameter constraints—exhaustively for finite domains such as `bool` and enum parameters, and as a bounded, deterministically sampled set (cap 256) otherwise—making it *bounded*, not complete. For every pure instruction it filters the state space by well-typedness and declared preconditions, executes the instruction, and evaluates three obligations per surviving state—progress (no raise where the precondition holds), preservation (declared outputs present and well-typed), and the postcondition itself. Side-effectful instructions are never executed and are reported out of scope; their obligation is delegated to authoring-time tests and review. Theorem 3.2 is conditional on this obligation, Theorems 3.4–3.6 carry the separate rewrite obligations of Appendix C.7, and beyond the enumeration bound the condition remains substantive rather than discharged (Appendices B.7 and C.7)—formally, the checker establishes $\forall s \in S_{bounded} \subseteq S : \text{pre}(i, s) \implies \text{post}(i, \text{exec}(i, s))$, where $S_{bounded}$ is the explored subset; completeness is traded for tractability—the standard bounded-exploration compromise, with the operational-semantics treatment of instruction meaning following [36].

The guarantee therefore comes in two explicitly separate layers. **Layer 1 (theory, conditional):** Theorem 3.2 and the type-safety results of Appendix B hold for every program whose instructions satisfy Definition 3.5 *on all of S*, under the total-definedness premise of Appendix B.7—a mathematical statement about a proof obligation, not about the checker. **Layer 2 (implemented verification):** the shipped checker discharges that obligation only on $S_{bounded}$ and only for pure instructions, as described above. Success of the bounded check is evidence toward, not a proof of, Layer 1's premise; no claim in this paper equates the two.

Theorem 3.4 (Optimization Soundness): Every atomic rewrite of the pass pipeline—fusion (Appendix C.2's rule, applied by the combining pass), dead-code elimination, and instruction combining (Section 3.4.4)—preserves the denotation of the instruction sequence it transforms: $\text{sem}(r(i)) \equiv \text{sem}(i)$, read as the partial equivalence of C.1.

Proof sketch: By structural induction on instruction composition. The base case shows each atomic rewrite preserves the denotation of a single instruction; the inductive step lifts preservation through each composition operator of Definition 3.6. The full proof, with the per-rewrite case analysis, is in Appendix C.

The shipped checker (`agentflow.core.isa.bounded_checker.verify_instruction`) implements Definition 3.5 on the bounded state space and reports violations structurally (listing in Appendix D.11 of the supplementary material).

3.4 Instruction Organization and Optimization

3.4.1 Instruction Composition. The AgentISA supports hierarchical instruction composition through a formal composition algebra:

Definition 3.6 (Instruction Composition): Given instructions $i_1, i_2 \in I$, we define four composition operators—Sequential $i_1 \circ i_2$, Parallel $i_1 \parallel i_2$, Conditional $i_1 \triangleright i_2$, and Iterative i_1^*—whose denotations (sequential execution; parallel interleaving under disjoint state footprints; guarded execution of i_2 on i_1's triggering outcome; zero-or-more iterations of i_1) are fixed in Appendix C.1.

Composition follows algebraic properties. Properties 1 and 3 hold for all instructions; Property 2 is a *conditional* law—purity and pairwise footprint disjointness together do *not* suffice, and idempotence of i_1 must be assumed separately:

1. **Associativity** (all instructions, up to denotation): $(i_1 \circ i_2) \circ i_3 = i_1 \circ (i_2 \circ i_3)$.
2. **Distributivity** (conditional on idempotent i_1 and pairwise-disjoint footprints): the law $i_1 \parallel (i_2 \circ i_3) = (i_1 \parallel i_2) \circ (i_1 \parallel i_3)$ holds under two premises that must be stated separately. **(a) Idempotence of** i_1: $\delta_{i_1} \circ \delta_{i_1} = \delta_{i_1}$ on the precondition-satisfying states—i_1's writes must not change the values its own subsequent execution reads (a constant overwrite of declared outputs qualifies; a read-modify-write of a cell it also reads does not). **(b) Pairwise footprint disjointness**: the denotations of i_1, i_2, i_3 act on pairwise disjoint state components (declared outputs distinct, read sets disjoint from the others' writes). Under (a) the second execution of i_1 reproduces the first's writes; under (b) the other instructions cannot perturb i_1's inputs between its two occurrences; disjoint updates commute, so both sides denote the same net transformation.

   Both premises are necessary, and neither follows from the other or from purity. Purity not implying idempotence is the sharper failure: a pure `update(+1)` applied twice increments twice (purity bars *external* side effects, not writes to declared cells). Taking i_1 as such an update and $i_2 = i_3 = \epsilon$ (the identity, empty footprint, so (b) holds vacuously), the two sides execute i_1 once and twice—from $x = 0$ they yield $x = 1$ and $x = 2$; only premise (a) rules this out. Conversely, without (b), a write of i_2 into a cell i_1 reads changes i_1's second execution even when i_1 is idempotent.
3. **Identity**: $i \circ \epsilon = \epsilon \circ i = i$.

   where ϵ is the empty (identity) instruction.

Remark (implementation consequence): distributivity also fails for side-effectful instructions, for the same double-execution reason. Because the law is doubly conditional, the shipped compiler does not rely on it at all: parallel grouping is gated on purity plus disjoint dependencies and disjoint outputs, `side_effects=True` instructions are excluded from all such rewrites, and no distributivity-based simplification is applied. The law is stated for the algebra's completeness, not as a license for rewrites.

3.4.2 Instruction Pipeline Design. We formalize the instruction pipeline as a directed acyclic graph (DAG):

Definition 3.7 (Pipeline DAG): $P = (V, E, \lambda, \delta)$, where V are the pipeline stages, E the dependencies between stages, λ the stage latency function, and δ the data transfer function.

The pipeline's stages (FETCH → DECODE → SCHEDULE → EXECUTE → COMMIT, with hazard detection, dependency tracking, and resource allocation maps; listings in Appendices D.2–D.3 of the supplementary material) are organized around this DAG.

Theorem 3.5 (Pipeline Correctness): For any valid instruction sequence s and any schedule S satisfying conditions S1-S4 of Appendix C.5—each instruction dispatched exactly once, dependencies respected, dispatch groups pairwise non-interfering, commit in program order—pipeline execution preserves program semantics.

Proof: By induction over the dispatch groups of S in commit order, under the invariant that the architectural state after committing each prefix of groups equals the sequential state after the same instructions; S1–S4 supply exactly-once dispatch, dependency order, group commutativity, and in-order commit (details in Appendix C.5).

3.4.3 Instruction Selection and Scheduling. Instruction selection follows a tree-pattern matching approach:

Definition 3.8 (Selection Pattern): $\pi = (\tau, C, \text{cost}, \text{code})$, where τ is a tree pattern over the IR (unrelated to the type-system symbol T of Definition 3.2), C the context constraints on the surrounding state, and cost/code the cost function and generated code.

Given an IR tree t to cover, the selection process is formalized as $\text{Select}(t) = \arg\min_{\pi \in \Pi} \{\text{cost}(\pi) \mid \text{matches}(\pi.\tau, t) \wedge \text{satisfies}(t, \pi.C)\}$.

Definition 3.8's optimal selection ships as the opt-in DP selector of Section 4.1.1; the default remains the heuristic matcher run by Section 5's benchmarks.

Scheduling is formalized as the classical delay-minimization problem under dependency and capacity constraints: minimize the total stall subject to dependency order ($\text{start}(i_2) \geq \text{end}(i_1)$ for every dependency pair) and to capacity at every time (formulation in Appendix E.1 of the supplementary material)—the specified, not shipped, scheduler's objective.

3.4.4 Static and Dynamic Optimization. **Static optimization.** Static optimization employs several techniques:

1. **Peephole Optimization**: $\text{PH}(i_1, \ldots, i_n)$ returns $\text{opt}(i_1, \ldots, i_n)$ if a matching pattern exists, otherwise the window is left unchanged.
2. **Dead Code Elimination**: $\text{Live}(i) = \text{Use}(i) \cup (\text{Live}_{\text{out}}(i) - \text{Def}(i))$. The two shipped passes approximate this fixpoint at different conservatism levels: the *compiler* pass (formal streams) removes a side-effect-free instruction when no live consumer depends on its id or any declared output, while the *trace optimizer* pass (executable instructions) keeps any instruction declaring at least one output—executable traces lack the formal dependency annotations that make output-consumption analysis exact. Section 5.3 exercises the former; Section 5.4 the latter.
3. **Instruction Combining** (the named candidate rules—textbook mnemonics, illustrative only, not shipped opcode names; listed in Appendix D.10 of the supplementary material—are specified and not registered: Section 4.1.3's only shipped write path into the rule set is the promotion gate of Section 3.3.1; the built-in pairwise compatibility combining is active, and it is what the hot-path trace re-optimization of Section 5.4 applies).

Dynamic optimization. Dynamic optimization in the shipped runtime is **repetition-triggered and threshold-based**; it does not (yet) respond to load or performance signals:

1. **Hot Path Detection** (implemented, two shipped modes): the default `exact` mode keeps a monotone counter per exact instruction sequence (`path_id` = the ordered join of instruction ids) and flags a path hot when its count exceeds a configurable threshold *and* its most recent recorded success rate is at least a stability bound (0.9 by default), $\text{Hot}(p) \iff \text{count}(p) > \theta \wedge \text{success}(p) \geq \alpha$—no windowing, no load signal.

   The opt-in `ema` mode instead triggers on recent execution share, over bounded per-path state (Section 4.2.2).
2. **Trace Optimization** (implemented): once hot, the cached sequence is re-optimized by the static pass machinery—peephole, dead-code elimination, instruction combining—bounded to a fixed number of passes; the runtime then serves the shrunken form for subsequent executions (demonstrated end-to-end in Section 5.8).
3. **Trace Formation and Adaptive Recompilation** (specified, not implemented): sub-trace extraction from observed frequency profiles, and performance-triggered recompilation (`should_recompile`: execution count above threshold *and* observed performance below expectation) remain design-level. The shipped runtime re-optimizes the whole cached sequence once per recording after the threshold is crossed, with no performance feedback loop.

Two consequences of the shipped design now have shipped remedies the default does not enable: exact keying means trace diversity (similar-but-not-identical sequences) never accumulates a shared count—the `ema` trigger relaxes the threshold to recent share but stays per-path keyed (residuals in Sections 4.2.2 and 6.1)—and hot-path state, formerly unbounded, is bounded by an LRU cap on tracked paths (`max_tracked_paths`, 1,000 by default) that evicts least-recently-recorded paths, hot ones included.

3.4.5 Verification and Analysis. Optimization correctness is argued through:

Theorem 3.6 (Pass-Pipeline Soundness, extending Theorem 3.4): For any optimization transform T composed of the pass pipeline's rewrites, $\forall p \in \text{Program} : \text{sem}(T(p)) \equiv \text{sem}(p)$.

Proof sketch: semantic preservation for each atomic transformation (the atomic-rewrite lemma, Appendix C.2) composes across transformations by the compositionality lemma (Appendix C.3); hazard-free pipeline execution is *not* part of this theorem—it is Theorem

3.5 (Appendix C.5), invoked separately; this theorem's contribution is the rewrite layer.

Performance Model (specified; the benchmarks of Section 5 measure latency and token counts directly and do not evaluate it): the modeled cost of a program sums, per instruction, base cost plus overhead minus an optimization gain that aggregates each optimization's benefit weighted by its application rate (both formulas in Appendix E.1 of the supplementary material).

3.5 Capability Assessment

Agents are heterogeneous: an agent may be strong at reasoning but weak at tool use, or vice versa. To make such differences explicit—and to give the multi-agent coordinator of Section 5.6 a principled basis for agent selection—AgentISA includes a capability assessment subsystem, implemented as `CapabilityProfile` in `agentflow.core.isa.capability`.

Definition 3.9 (Capability Profile): A capability profile of an agent a maps each capability domain D (the shipped enum carries eight domains: reasoning, memory, generation, optimization, specialization, coordination, security, robustness) to a score–weight pair, $\mathrm{Cap}(a) = \{(\mathrm{dom}_k, s_k, w_k) \mid \mathrm{dom}_k \in D\}$:

where each domain carries a score $s_k \in [0, 1]$ and a weight $w_k \geq 0$ reflecting the domain's importance in the current deployment context.

Implementation status. All three layers ship: the aggregation and selection mechanisms (`CapabilityProfile`, `get_overall_score`, and the coordinator's threshold selection) and—replacing the former stub—the instruction-level metric extraction itself. The seventeen `_measure_*` functions of `agentflow.core.isa.capability` now extract real signals statically from `FormalInstruction` *declared metadata*: prevalence of category labels, dependency edges, parameter schemas, pre/postconditions, and declared outputs; working-memory complexity as the largest declared input arity (saturating at four inputs); and resource economy as the inverse of the declared resource/cost footprint. Extraction is deterministic, side-effect-free, and conservative: absent declarations score zero. The pipeline's *mechanics* are pinned by unit tests (`tests/core/isa/test_capability_metrics.py`, `test_capability_pipeline.py`), including the double-weighting guard disclosed below. One wiring gap remains, disclosed: `assess_capabilities` still has no production call site—the coordinator's selection path receives injected `DomainScore` values (Section 5.6's tests inject them directly).

Assessment proceeds in three steps:

1. **Domain metric computation.** For each domain, `assess_capabilities` computes a set of `CapabilityMetric` values from the agent's instruction set by static extraction, reading declared metadata only: no execution, no live probing.
2. **Domain aggregation.** Metrics are aggregated into a per-domain `DomainScore` as a confidence-weighted mean, matching the shipped `_compute_weighted_score`: $s_k = \frac{\sum_{m \in M_k} v_m \cdot \mathrm{conf}_m}{\sum_{m \in M_k} \mathrm{conf}_m}$, where M_k is the metric set of domain k, v_m the metric value, and conf_m its confidence; an empty metric set scores 0, and zero total confidence degenerates to the unweighted mean. The domain weight w_k is deliberately absent here—it enters only once, at the top level (an earlier form folded it into s_k as well—double-counting; Section 5.6's injected-score tests were unaffected).
3. **Overall score.** `get_overall_score` computes the weighted mean across assessed domains, $S(a) = \frac{\sum_k w_k s_k}{\sum_k w_k}$, falling back to the unweighted mean when all weights are zero.

Three properties of this design deserve emphasis. First, it is **static rather than live-probing** (runtime statistics and cross-instruction dataflow-consistency checks remain outside its scope). Second, it is **compositional**: fusing or eliminating instructions during compilation (Section 3.4) changes the profile predictably, and a production deployment would re-assess after optimization (the shipped coordinator does not yet do so). Third, it directly serves coordination: the agent-selection predicate in Section 5.6 chooses the highest-scoring agent above a capability threshold for each task, subject to the security policies of the `SecurityMonitor`—whose exercised selection path still receives injected scores, as disclosed above.

We state the scope of this subsystem honestly: `CapabilityProfile` assesses *declared instruction-level capability*, which correlates with—but does not measure—end-to-end task performance. Cross-model behavioral comparison at benchmark level is the domain of evaluation suites such as BFCL [17] and AgentBench [29]; the two views are complementary.

4 Implementation

This section describes the implementation status of Section 3's components, with emphasis on the compiler infrastructure and optimization techniques of Section 3.4.

4.1 Compiler Infrastructure

4.1.1 Instruction Selection Implementation. We implement the tree-pattern matching approach of Section 3.4.3 (Definition 3.8) as dynamic programming over the IR tree, in the classical manner of bottom-up instruction selection [33]. The MATCH-TREE selector ships in `agentflow.core.isa.instruction_selector` as an *opt-in strategy* (`selection_strategy="dp"`, engaging only when the input carries an `ir_tree`; a catalogue instruction declares a multi-node pattern via `params["spanned_child_ops"]`), following the Appendix D.1 recurrence exactly and exposing the DP table and root-first covering of Theorem 4.1. The *default* (`selection_strategy="heuristic"`) remains the name-overlap heuristic matcher, unchanged—so every benchmark number in Section 5 is produced under the default and none depends on tree-pattern selection.

Theorem 4.1 (Selection Optimality, shipped opt-in selector): Given a set of context-local patterns P and an IR tree T, the MATCH-TREE algorithm computes the minimal covering cost; the covering itself is read off the `selected` table afterwards, by walking from the root and skipping subtrees spanned by the chosen pattern.

Proof sketch: by induction on tree height; the recurrence sums only over children not covered by the candidate pattern, so minima compose (full argument, with the context-locality side condition, in Appendix D.1). This theorem is a property of the shipped opt-in selector, not of the default heuristic; no optimality claim is made for the default path.

4.1.2 Pipeline Scheduling Implementation. We *specify* the pipeline scheduler following the instruction-scheduling formulation of Section 3.4.3 (Deps and resource capacity) and the correctness conditions of Theorem 3.5 (Section 3.4.2), with hazard detection and resource-constrained issue in the style of classical pipeline scheduling [34]. **This cycle-accurate scheduler is design-level**: the executed runtime path is the sequential and parallel-grouped dispatch produced by the compiler's grouping pass (Section 4.1.3); no benchmark in Section 5 exercises a cycle-accurate schedule. The specified scheduler is listed in Appendix D.2.

Schedule verification checks the dependency-ordering condition (S2) of the formal model (Theorem 3.5, Appendix C.5), with the edge convention of Section 3.4.3—$(i_1, i_2) \in \text{Deps}$ means i_1 must finish before i_2 starts. The check as written, $\forall (i_1, i_2) \in \text{Deps} : \text{slot}(i_2) > \text{slot}(i_1)$, is equivalent to $\textit{start}(i_2) \geq \textit{end}(i_1)$ only under the **unit-slot model** the shipped scheduler assumes—one slot per cycle, every instruction occupying exactly one slot. Admitting multi-cycle latencies would require an explicit *start*/*end* comparison (or interval slots); that extension remains the specified component of Section 4.1.2. S1 (exactly-once dispatch) and S4 (in-order commit) hold by construction of the scheduling loop, and S3 (group non-interference) is enforced by the grouping check.

4.1.3 Static Optimization Implementation. We implement the optimization passes defined in Section 3.4.4. The passes that fire in the benchmarks are fusion (the instruction-combining pass of Section 3.4.4, listed in Appendix D.10 of the supplementary material), dead-code elimination (the shipped two-level design of Section 3.4.4), and parallel grouping (described with its measurement in Section 5.3). The shipped rule set registers no peephole patterns by default, so every peephole obligation below is dormant in the shipped configuration but stated for completeness (one shipped write path can populate it: approved mining-to-optimization proposals, Section 3.3.1, install name-level rules through the audited gate):

1. **Peephole Optimization** (shipped class with an empty pattern table; listing in Appendix D.8):

Each peephole rule carries a correctness *obligation*, discharged as part of rule definition rather than by the running compiler: every pattern must satisfy $\text{matches}(p, w) \implies \text{sem}(\text{replace}(p, w)) \equiv \text{sem}(w)$ for all instruction windows w.

As stated above, the shipped rule set is empty in the default configuration, so the equivalence checker is dormant; Section 4.3.1 specifies the symbolic-execution checker a populated rule set would require (the promotion-gate rules are name-level rewrites, not the textbook patterns below).

2. **Instruction Combining**: the shipped pairing pass groups adjacent compatible instructions and verifies each merged group; its full listing is in Appendix D.10 of the supplementary material, and in the source module `agentflow.core.isa.instruction_optimization`.

4.2 Runtime System

4.2.1 Dynamic Pipeline Control. Design of the dynamic pipeline controller (**specified, not implemented**: the shipped dynamic machinery is the hot-path detector (two modes; Section 4.2.2) and the trace re-optimization of Section 3.4.4, measured in Sections 5.4 and 5.8; a skeleton module exists but is not wired into the runtime, Appendix D.3). Pipeline performance is modeled with the classical pipeline bound, $\text{Throughput} = 1/\max_i \text{service_time}(\text{stage}_i)$—a model of the specified controller; nothing in Section 5 measures it.

4.2.2 Dynamic Optimization Implementation. The *default* mechanism is the counter/threshold detector described in Section 3.4.4: a monotone count per exact `path_id`, crossed against a configurable threshold with a last-recorded-success-rate gate. What the benchmarks measure (Sections 5.4 and 5.8) is this default. The Appendix D.4 detector—exponential moving average over a window, frequency normalized by recent execution share—now also ships, as the opt-in `detection_mode="ema"` of `agentflow.core.isa.instruction_optimization` (details in Appendix D.4). It addresses workloads with trace diversity at the trigger level only: the detector itself remains exact-path-keyed (the residual of Section 3.4.4—similar but non-identical sequences never share a counter—still applies), and the share denominator is the LRU-tracked set, so a long tail of one-shot paths can keep a later burst below threshold until the cap evicts them. Block-granular trace formation below remains specified, not implemented:

1. **Hot Path Detection** (shipped opt-in; listing in Appendix D.4):

The decision rule of the pseudocode is the threshold predicate IS-PATH-HOT (frequency above θ, stability above the stability threshold); it computes no score. A *ranking variant*—specified, not part of the decision rule—orders candidate paths by a heuristic β-weighted blend of the already-normalized frequency and recent stability, not a conditional probability.

2. **Trace Formation** (specified; listing and validity definition in Appendix D.5): trace validity is the conjunction of three properties—entry dominated by the loop header, exit post-dominating the tail, no side exit except the tail—checked by VERIFY-TRACE-PROPERTIES.

4.3 Optimization Verification

4.3.1 Static Analysis. We *specify* verification of the static optimizations from Section 3.4.4 (a symbolic-execution equivalence checker over transformation path pairs). As stated in Section 4.1.3, the shipped peephole table is empty and this obligation is dormant; the soundness theorems (Theorems 3.4–3.6) are conditional on discharging it when rules are introduced (Appendix C.7). The verification follows the formal model, $\text{Equiv}(p_1, p_2) \iff \forall \sigma \in \Sigma : [\![p_1]\!]\sigma = [\![p_2]\!]\sigma$; the specified checker is listed in Appendix D.6.

4.3.2 Dynamic Analysis. The runtime *monitors* optimizations through a single mechanism (no result-correctness oracle is defined at runtime—correctness of LLM-instruction results is not decidable in general and is delegated to the authoring-time verification of Section 3.3.3; the monitor below is a specified, not shipped, component):

1. **Performance Monitoring** (specified; listing in Appendix D.7).

4.3.3 Empirical Results. The implemented optimizations are validated by the microbenchmark suite of Section 5, which drives the real compiler, cache, and optimization-manager code paths against a mock LLM backend. We keep all numbers in Section 5; in map form: result caching is validated in Section 5.2, the fusion and DCE static passes in Section 5.3, hot-path detection and adoption in Section 5.4, per-pass attribution in Section 5.5, coordination and security in Section 5.6 (43 functional tests), the live-gateway cache and compiled pipeline in Sections 5.7–5.7.1, and whole-system composition in Section 5.8—every measurement a controlled-setting result on synthetic or otherwise disclosed workloads.

5 Empirical Evaluation

We evaluate AgentISA with a reproducible microbenchmark suite (`examples/benchmarks/isa_microbench.py`) and a multi-agent functional test suite, under the scope fixed at the end of Section 4.3.

5.1 Methodology

5.1.1 Instrumentation: Mock LLM Backend. All experiments route model inference through a `MockLLMBackend` that counts each `infer` call as one simulated LLM API round trip and charges a fixed 80 ms simulated latency per call to a simulated clock. It never sleeps and never contacts a live model, so every "API call" is observable, deterministic given a seed, and reproducible without network access or inference spend. Everything *around* the backend is real: the cache, compiler, and optimizer are the production `agentflow.core.isa` implementations (`InstructionCache`, `InstructionCompiler`, `OptimizationManager`), exercised through the same code paths a deployment uses.

5.1.2 Workloads. Workloads are synthetic and seed-controlled:

- **Caching**: 2,000 tasks from a discrete distribution over 50 instruction templates × 20 parameter values, popularity Zipf-distributed with exponent $s \in \{0.5, 1.0, 1.5\}$. Three arms share the same instrumentation (Section 5.2): no cache, the `InstructionCache`, and a naive app-level exact-response cache (a plain dict keyed on the request payload).
- **Static compilation**: random sequences of length 50, 200, and 1,000 from `synthetic_formal_sequence` (`examples/benchmarks/isa_microbench.py`; short enough to audit line by line). Each of the first $n-1$ slots is drawn independently: with probability 0.25 a *dead* instruction (side-effect-free, **no declared outputs, hence no consumers anywhere in the sequence**—removable by construction, planted not inferred); with probability 0.55 a *pure* instruction (side-effect-free, one declared output v_i, consumed **only by the final sink**, fusable next to a same-type pure neighbor); otherwise *side-effectful* (always live, never fused or removed). The n-th slot is a side-effectful *final sink* depending on exactly the set of all pure outputs, so every pure instruction has a downstream consumer and no pure output is deleted as dead. The attributes are disjoint and jointly exhaustive per slot; the passes find the planted DCE/fusion targets, so the measured reductions are properties of this workload family, not of arbitrary agent programs.
- **Dynamic optimization**: instruction traces of length 100 recorded for up to 200 iterations per seed, hot-path detection active (threshold 50 recordings); the loop terminates at the trigger—iteration 50 in all seeds—so each seed contributes exactly 51 recordings (the 50 counted plus the trigger).
- **Ablations**: two arms. The primary arm is *paired*: it recompiles the static-compilation sequences (identical seed base 2,000) with fusion and dead-code-elimination toggled individually and jointly—a same-input decomposition of the 34.2% headline. The secondary arm is an *independent* draw, same distribution but seed base 4,000, a fresh-sequence replication (all-passes total 35.0%, within one standard deviation of the headline).

5.1.3 Protocol and Statistics. Each synthetic-workload configuration (Sections 5.2–5.5) is run with 5 random seeds; we report mean ± sample standard deviation. These are **descriptive statistics only**: no hypothesis-test claims (no p-values, no significance tests) anywhere in this paper. Five seeds characterize variability across the tested workload instances, not statistical significance or population-level generality. Section 5.2's naive response-cache baseline shares the mock-backend instrumentation; a formal *inferential* comparison against full external systems remains out of scope, and no such claim is made.

5.1.4 Metric Terminology. Several distinct quantities must not be conflated; each is defined once here:

- **Effective-instruction reduction**—fewer *effective* instructions in the compiled program (Sections 5.3/5.5); an instruction-stream property, not execution time.
- **Dispatch-count reduction**—fewer dispatch operations at runtime, including parallel grouping's folding (Section 5.3); a runtime count.
- **Trace shrinkage**—shorter traces after hot-path reoptimization (Section 5.4); the optimized trace's executed-state identity is verified separately there.
- **Cache hit rate / call reduction**—fraction of calls served from the result cache; here the call-reduction ratio equals the hit rate by construction (Section 5.2).
- **Simulated latency saving**—avoided-call counts × the mock's fixed 80 ms (Section 5.2); derived, not measured.
- **Measured latency/token saving**—wall-clock time and token counts against the real gateway (Sections 5.7–5.7.1); measured.

For context preservation (Theorem 3.3): *extraction-time preservation* is the extraction step's copy invariance; *runtime context validation* is the monitor's pre/post-dispatch constraint checking—different properties, different coverage (Section 3.2.2 discloses the deletion blind spot).

5.2 Experiment 1: Result Caching

Research question. How many simulated LLM API calls does AgentISA's result caching eliminate, as a function of workload skew?

We execute 2,000 tasks per seed, with caching disabled and with the `InstructionCache` enabled, measuring simulated API calls at the backend; the reduction ratio and hit rate coincide by construction (every avoided call is exactly one cache hit).

Cache-key contract. The shipped result cache keys an entry on the (instruction, params) pair—realized as a params-only SHA-256 scoped per instruction instance—deliberately *not* on model identity/version, prompt-template version, tool schema, decoding parameters, or ambient context. Soundness requires those factors held fixed, which every benchmark here guarantees by construction (single alias, fixed templates and decoding, fresh cache per arm); Section 5.7.1's classifier-keyed step hashes exactly the input dictionary its prompt is built from, under the same discipline. A deployment changing any of these factors must invalidate or namespace the cache. The production cache isolates by TTL, LRU capacity, and never-cached failures, and an opt-in strict namespace now ships for this drift (`CacheKeyNamespace` in `agentflow.core.isa.instruction_cache`, per instruction via the optional `cache_namespace` attribute): the key extends (instruction, params) with model alias, template version, and a canonicalized decoding hash (benign key reorders do not over-invalidate), with isolation-not-eviction semantics—rollback re-hits the resident entry—while the default key stays the params-only SHA-256, byte-identical to the published hit statistics. *Automatic* drift detection remains absent, and no number in this paper runs under the strict key.

Naive response-cache baseline. A third arm runs an app-level *response* cache keyed on the exact request payload string—what an exact-match cache in front of the backend (GPTCache's exact mode, or an HTTP response cache) keys on—implemented as a plain dict with no AgentISA machinery. Across the three skews it issues exactly the same call counts as the table's cached arm: **parity with the `InstructionCache` to the digit**, because the (instruction, params) key is already canonical on this workload. Parity is the finding, stated plainly: at the level of call reduction the instruction-level cache adds nothing over exact-response caching; what it adds is the compile-time transformations of Sections 5.3 and 5.5—fusion and dead-code elimination, reducing the program before any request is issued, which a response cache cannot express because it sees request strings, not instruction semantics. Semantic/embedding caching remains out of scope (Section 5.9).

Interpretation. The reduction grows monotonically with skew, 59.5% at $s = 0.5$ to 79.6% at $s = 1.5$: the savings' upper bound is the probability mass of repeated tasks, which the Zipf exponent directly controls. In simulated latency the savings are the avoided-call counts × 80 ms ($s = 0.5/1.0/1.5$: 95,168 ± 455 / 108,640 ± 305 / 127,360 ± 750 ms)—derived, not separately measured.

Two caveats keep this honest. First, these are *simulated* calls against a mock backend: the reduction is exact for this workload family, but real deployments face cache-invalidation, prompt-context, and embedding-variability effects the mock abstracts away. Second, the absolute number depends on reuse structure; workloads without repetition would see zero benefit. The experiment establishes the *mechanism and its skew sensitivity*, not a universal constant.

5.3 Experiment 2: Static Compilation

Research question. How much does the compiler reduce instruction sequences, and what do the different optimization levels contribute?

We compile random sequences at four optimization levels and measure two quantities, kept explicit (they are easy to conflate): **effective instruction count**—instructions remaining after fusion and dead-code elimination, a program-size measure—and **dispatch count**—independent dispatch operations the runtime issues, where parallel grouping can fold several independent instructions into one dispatch *without changing the effective instruction count.*

Interpretation (full table: supplementary I.3). Three observations:

1. **Effective instruction count reduction is level-stable**: BASIC, ADVANCED, and AGGRESSIVE produce identical reductions (34.2% ± 1.8% at length 200). BASIC's fusion and DCE already capture the available elimination gains; higher levels add *scheduling* improvements (dispatch folding), not elimination—ADVANCED and AGGRESSIVE coincide here, their extra passes finding no further targets on these sequences, differing only on streams carrying combinable patterns. We report this as-is rather than claim monotonic gains.

2. **Dispatch count drops further with parallel grouping**: at length 200, dispatch falls from 131.6 (BASIC) to 74.0 (ADVANCED) with effective-instruction reduction unchanged—adjacent side-effect-free instructions with disjoint dependencies and outputs grouped into shared dispatches. The 74.0 figure is a scheduling reduction, not a further program-size reduction. (An earlier implementation folded more aggressively—44.4 at length 200—but could place side-effectful instructions into shared groups whose non-interference was never established; the sounder grouping here folds less, and Appendix C.5's dispatch-folding corollary now matches the implementation exactly.)

3. **Gains are stable across sequence lengths**: 32.8%, 34.2%, 35.4% at lengths 50, 200, 1,000 respectively, tightest variance at length 1,000 (± 1.6%); length does not materially change the opportunity structure.

Effective-instruction reduction is again a *program-size* metric: it translates to simulated API-call savings only where eliminated or fused instructions carry model inference; the caching experiment measures the complementary reuse effect.

Table 1: Hot-path trace shrinkage per seed (length-100 traces, detection threshold 50 recordings). The replay column records the executed-state identity check: the optimized trace, re-executed against an independent state, reproduces the original trace's final state exactly in every seed.

Seed	Original length	Optimized length	Shrink ratio	Replay state identical
0	100	30	70.0%	yes
1	100	41	59.0%	yes
2	100	30	70.0%	yes
3	100	36	64.0%	yes
4	100	34	66.0%	yes

5.4 Experiment 3: Dynamic Hot-Path Optimization

Research question. Does the dynamic optimizer detect and specialize repeatedly executed traces?

We record an instruction trace of length 100 for up to 200 iterations per seed with hot-path detection active (Section 5.1.2). Once a path is flagged hot, the runtime forms an optimized trace; **this experiment measures that trace's length and re-executes it**—the replay verification below checks that it reproduces the original trace's final state exactly (task-level artifact preservation is established separately, on realistic streams, by Section 5.8).

Across seeds, hot-path optimization shrinks the trace by **65.8% ± 4.6%**; the seed-to-seed spread (59.0%–70.0%) reflects which subsequences each synthetic trace makes hot, the effect present in all five seeds. As with the static experiments, this is trace-length reduction on synthetic traces—workload-controlled evidence that the detection-formation-reoptimization loop functions, not a wall-clock speedup claim.

Replay verification. Each seed's optimized trace *is* re-executed—every instruction reads its declared dependencies from a threaded state dict and returns deterministic bindings for its declared outputs (side-effectful instructions leave per-id markers)—and its final state is compared with the original's, executed the same way. The states are identical in all five seeds (5/5; recorded per seed as `replay_state_identical` in the results file), so the 65.8% shrinkage carries an executed-equality check at trace level—on the deterministic mock, not against a live model; this is the same style of check as Section 5.8's task-level byte-identity, applied to the synthetic traces. Live-model behavior remains untested—no mock experiment can address it.

5.5 Ablation: Per-pass Contribution

To attribute the static gains we run the two ablations of Section 5.1.2. The primary one is **paired**: it compiles the *same* sequences as Section 5.3 (identical seed base 2,000 and generator) with fusion and DCE toggled individually and jointly, so per-configuration differences are same-input paired differences—an exact decomposition of the headline, not a replication:

The paired per-pass decomposition (table: supplementary I.3) gives fusion+DCE = 34.2% ± 1.8%—the headline itself, confirming that these two passes account for the entire static gain at length 200. The **paired interaction**, computed per seed as (fusion+DCE − NONE) − (fusion − NONE) − (DCE − NONE)—34.2 − 21.6 − 24.5 = −11.9 points—is −11.9% ± 2.0% (per-seed −14.5/−9.5/−13.0/−12.0/−10.5 points): the passes overlap—dead slots break fusion chains and fused pairs absorb neighbors DCE would otherwise have removed individually—so the sum of the single-pass gains (46.1 points) overstates the joint effect by 11.9 points.

The secondary ablation is an **independent-draw replication** (seed base 4,000, same distribution): fusion alone 21.9% ± 3.5%, DCE alone 24.1% ± 3.3%, fusion+DCE and all passes 35.0% ± 3.7%—the paired structure reproduces on fresh sequences, so the decomposition is not an artifact of one draw. "All passes" equals fusion+DCE here, consistent with Section 5.3's level stability: no additional pass contributes measurable reduction, and we report that absence rather than dress the higher levels in unearned credit. Because dead instructions are planted at rate 0.25 by construction, DCE's per-pass share is near-tautological for this family—it measures that the pass finds the planted dead code, precisely its function.

5.6 Multi-Agent Coordination: Functional Validation

The multi-agent layer—`AgentCoordinator`, `SecurityMonitor`, and the collaborative instruction set—is validated functionally by 43 tests, all passing—the coordination-and-security subset of the repository's full suite (1,333 passing, 14 skipped at run time), which also covers the ISA core, compiler, and optimizer units of Sections 5.2–5.5. This is **qualitative functional validation**: the tests verify the coordination behaviors exist and are correct, not that they are fast; no performance number should be inferred here.

The coordination flow implemented by `AgentCoordinator` is listed as pseudocode in Appendix D.10 of the supplementary material. What the 43 tests verify, grouped by subsystem:

1. **Agent registration and role policies**: registration creates policies; permissions and security requirements differ by role; trust levels are bounded and reward declared specializations.
2. **Task coordination**: coordination produces a solution or records failure; missing authentication/authorization and no-suitable-agents raise; selection respects the capability threshold, the coordinator being the highest-scoring agent.
3. **Execution planning**: stage structure with coordination rules; every stage has differentiated fallback keys and a fallback executes on stage failure; validators are reasoning-heavy non-coordinator agents.
4. **SecurityMonitor**: valid tasks pass, missing authentication/authorization fails, violations are timestamped, declared data sensitivity is enforced (sensitive categories infer a high sensitivity level, benign ones pass at medium), resource access is open by default and restrictable by configuration.

5. **Collaborative instructions**: barriers wait for the declared participant count and time out on a missing peer; resource allocation rejects over-allocation and serializes concurrent requests; the workflow state machine accepts valid actions and rejects invalid ones and double starts; a research-team scenario runs end to end.

These tests exercise the same production modules used elsewhere and are fully deterministic (no LLM calls). They establish correctness of the coordination machinery under the tested conditions; what they do not establish is scalability under contention, crash recovery, or adversarial inputs beyond the policy checks above—all future work (Section 6).

5.7 Real-LLM End-to-End Validation of Result Caching

Research question. Does the caching mechanism behave identically when the "model" is a real LLM behind a live gateway, and how much *measured* wall-clock time and token usage does it save?

Setup. We replace the mock backend with a real model behind a local Anthropic-protocol gateway (TLS with an internal CA; alias `claude-glm-5.3-flash`, a hosted reasoning-capable GLM model; `max_tokens=64`; sequential httpx, 90 s timeout, three bounded retries—none occurred). Every other component is the production code path: the same `InstructionCache` protocol, instruction objects, and workload generator, with Section 5.2's instruction pool (50 templates) and parameter space (20 values).

Scale. With each round trip costing ~2.2 s, scale drops from 2,000 tasks × 5 seeds to **200 tasks × 1 seed** per Zipf level (s in {0.5, 1.0, 1.5}): 1,040 measured round trips (plus one preflight) in 39.4 minutes. The seed pins the task sequence exactly, so the workload is the one the expectation below is computed from.

An exactness property. A cache hit depends only on the (instruction, params) pairs seen so far—not on what the model returns. In general an LRU-with-TTL hit also depends on wall-clock time and capacity; under this benchmark none of that can bind: `max_size` = 100,000 (at most 200 distinct keys per arm), `ttl` = 3,600 s (every cache-enabled arm completes in under seven minutes; the no-cache arms run to 7.9 minutes and never consult a cache), a fresh cache per arm, failed calls never cached. Hit/miss decisions are therefore a pure function of the (instruction, params) sequence, *computable without any backend*; if the mechanism is correct the real run must reproduce that sequence exactly—and at the recorded granularity (per-arm hit rates and call counts) it does:

Per-call latency was stable across arms (p50 2.13–2.21 s, p95 3.06–3.54 s). Token savings are gateway-reported `usage` differences: every avoided call saves its input tokens plus the tokens the reasoning model generates before responding (output tokens were the binding constraint of `max_tokens=64` in every call).

Interpretation. First, the mechanism transfers unchanged: observed hit rates match the computed expectation exactly at every skew level, so Section 5.2's synthetic results measure the cache, not a mock artifact. Second, avoided calls convert into measured time and tokens—14.1%, 26.6%, and 45.8% of the no-cache arm's wall clock at s = 0.5, 1.0, 1.5—tracking but not identical to the call reduction (the exact quantity is the call reduction; wall and token figures are measurements). Third, the absolute hit rates (8.5–45.5%) trail Section 5.2's (59.5–79.6%) purely because repetition accumulates with task count: at N = 2,000 the same computation yields the Section 5.2 numbers.

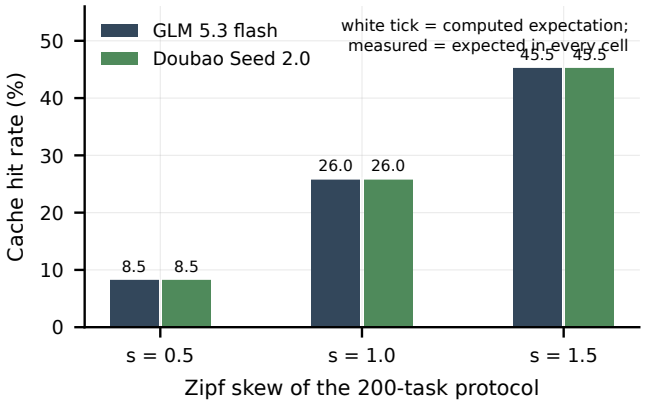


Figure 2: Real-gateway cache exactness across model families. Bars are measured hit rates under the identical protocol (200 sequential requests, one seed, three skews per model); the white tick is the backend-free computed expectation—matched exactly in every cell of every family; call, wall-clock, and token figures: supplementary I.3.

Multi-model rerun. The identical protocol re-run on a second production model (`claude-doubao-seed-2.0-code`, replies completing below the ceiling) reproduces the expectation exactly in all three cells—8.5%, 26.0%, 45.5%; calls 200 → 183/148/109; p50 3.6–4.0 s (table: supplementary I.3)—the exactness property is model-independent by construction.

Two-seed rerun. A reduced-scale rerun (100 tasks × 2 seeds per Zipf level) reproduced the computed expectation exactly in all six (Zipf, seed) cells (hit rates, quota record: supplementary I.3).

Scope of the claim. This experiment validates the caching mechanism end-to-end—one gateway, two production models (one seed per model), N = 200, sequential requests, no streaming—recording per-arm aggregate hit rates and call counts; per-position identity follows from the deterministic cache decisions above but is not separately logged, and nothing here is a throughput, availability, or dollar-cost claim. Sections 5.3–5.5 are structural, never dispatching to the backend; Section 5.7.1 extends the real-gateway validation to them via a cache-mediated protocol.

5.7.1 Real-Gateway Validation of Compilation and Hot-Path Optimization. **Research question.** Do static compilation and closed-loop hot-path adoption preserve the observable semantics of an instruction stream when the stream's LLM step is served by a real model—and what do they measurably save on a live gateway?

Setup, arms, scale. A dedicated harness (`examples/paper_real_llm_compiler_test.py`; tests in `tests/system/test_real_llm_compiler_test.py`) reuses Section 5.7's gateway configuration

(alias `claude-glm-5.3-flash`, sequential httpx, bounded retries) but raises `max_tokens` to 512 (Section 5.7 used 64) so completions dominate the priced comparison; it runs identically online or offline against a deterministic same-protocol mock (call counting, template-accounted tokens). The task is Section 5.8's service-log triage shape: a seven-instruction stream with one cacheable LLM-type step (`classify_signature`, one model call per distinct incident-signature batch) and the same planted dead guard (`check_format`), over 50 realizations × 1 seed × Zipf s = 1.0 from a fixed universe of 40 task instances—22 distinct signature batches, so the entire run is 23 gateway round trips in 4.4 minutes—bounded validation, not the 200 × 3-skew sweep of Section 5.7. Arm (a) is the unoptimized authored stream, run first, cold, performing every live round trip; arm (b) drives the *same* stream through the production machinery—`optimize_hot_path` adoption plus static `InstructionCompiler` at NONE vs AGGRESSIVE with the dataflow-preservation assertion—then executes the same draws against arm (a)'s warmed cache. The classify step's cache key is the sha256 of exactly the input dictionary its prompt is built from, so arm (b) issuing *zero* new gateway calls proves it computed identical keys—hence identical prompts and consumed values—and its artifacts must come out byte-for-byte equal.

Results. All checks pass. Arm (a) made exactly the 22 first-occurrence calls the backend-free computation predicts and served the other 28 realizations from the result cache (hit rate 56.0%), consuming 4,027 prompt and 11,264 gateway-reported completion tokens in 260.2 s of measured call latency. Arm (b)—compiled, hot-path-adopted—made zero new gateway calls and reproduced every artifact byte-for-byte. The hot path was adopted at realization 7 (threshold 5; adoption lands two realizations past it because the counter must strictly exceed it and recording follows execution, as in Section 5.8): realizations 1–6 executed the full seven-instruction stream, realizations 7–50 the shrunken six-instruction form with `check_format` absorbed into its successor by fusion—306 instruction executions against 350 in arm (a), a 12.6% executed-instruction reduction attributable to the adopted trace (per-stream shrink 14.3%, 7 → 6; realized saving diluted by the six full-form realizations before adoption; the static compiler alone reproduces the 7 → 6 shrink at AGGRESSIVE, so the two mechanisms' contributions are not separated), reported separately from the cache's call savings. Static compilation of the same stream: NONE 7 → AGGRESSIVE 6 effective instructions, dataflow preserved.

Identity rerun with live-model values. Re-running the harness on `claude-doubao-seed-2.0-code` at `max_tokens` = 1,500 changes the evidential class: all 104 live calls across the three Zipf cells completed below the ceiling (max completion 142 tokens, 0 bound) and replied in the requested reply format, so the artifacts' severity values are the live model's own classifications, not the fallback. Every mechanism check re-passed—expected call counts, arm (b) zero new calls, byte-identical artifacts, hot-path adoption, the 7 → 6 shrink—and a third model (`claude-minimax-m3`, reasoning-style) passes every check while binding at the ceiling in 97% of calls at `max_tokens` = 512 and 65% at 1,500: ceiling-binding is a model-family property, not a harness artifact (census: supplementary I.3).

A priced comparison (summary; protocol and table in Appendix G of the supplementary material). Costs are MODELED on measured tokens, not observed: the overlay prices this run under no optimization, AgentISA result caching (first occurrences only), and a modeled provider prompt cache (repeat prompts bill as 0.1x cache reads, mirroring Anthropic's published discount; completions at full price); two simplifications favor the provider-cache baseline, and list prices are parameterized stand-ins—no dollar-savings claim is made. On this run's token mix the result cache cuts modeled cost by 56.0% while the prompt cache cuts only 3.3% (the reasoning model's completions dominate); on the mock at paper scale the corresponding figures are 80.1%, 81.6%, and 85.1% against 28.9–31.4%—the ranking of the two caching strategies is a function of the input/output token mix, not a constant. GPTCache is explicitly out of scope; no head-to-head is claimed.

What this does and does not show. The byte-identity property is cache-mediated by design: the values entered the artifacts once through arm (a)—fallback-derived on this alias, live-model classifications in the doubao rerun—and arm (b) is proven to compute the same keys and consume the same values; this validates the stream transformation, not model determinism, which is not guaranteed at temperature 0 and is not claimed. On this alias the reasoning model's replies bind at `max_tokens` in every call (as at `max_tokens=64` in Section 5.7), so the classify step's severity values come from its deterministic fallback; what is validated against the live model is the mechanism chain (key computation, cache decisions, call and token accounting, artifact identity), not classification quality.

What the real-gateway experiments establish, in three layers kept separate: (1) *gateway integration correctness* (requests reach the gateway, the cache governs calls, call/token statistics recorded faithfully) and (2) *workflow semantic preservation* (the optimized stream computes the same cache keys and consumes the same cached values, yielding byte-identical artifacts—as a cache-mediated identity validating the stream transformation, not model determinism): **established** by Sections 5.7 and 5.7.1. (3) *Real-model task quality*: **measured only by a follow-up rubric diagnostic**—a 3,072-token rerun of the same 22 signature batches parsed 144 of 176 claims, with final classifications matching the planted keyword-rubric gold on 45.5% of signatures (model claims 51.4%) against 32.4% for the deterministic fallback rule, so the live model measurably outperforms the fallback rule on this rubric; 4 of 22 batches still bind at the ceiling, and the gold is a synthetic rubric, not human annotation—a rubric-following diagnostic at one gateway and reduced scale, not a deployment-quality claim (coverage limits restated with Section 5.9). A second quality run on a completing model (`claude-deepseek-v4-flash`, 36 distinct signature batches at the paper preset, `max_tokens` 1,500) parses 288/288 claims with 0 ceiling bindings: final classifications match gold on 52.1% of signatures against 31.9% for the fallback rule (per-class agreement 67.6/39.0/47.8/54.4% on low/medium/high/critical)—model-over-rule holds on both families (protocol: supplementary I.3).

5.8 Controlled Four-Task System-Level Characterization

The experiments above measure individual mechanisms on synthetic sequences; this section is a *controlled four-task system-level characterization*—a functional demonstration under a fixed closed-loop protocol, not a generalization experiment; the identical 91.7% figure below is a property of that protocol. The seed-controlled characterization (`examples/system_task_workload.py`, `tests/system/test_system_task_workload.py`) is built from **four tasks with genuinely different dependency shapes**—linear service-log triage, fan-out csv-etl-anomaly, wide-fan-in report-merge, two-branch regression-diff—each realized end-to-end against sandbox-resident files (12 realizations each, hot-path threshold 5, one planted dead guard and one cacheable LLM-type step per task, backed by a deterministic rule stand-in), making the mechanisms' composition and Section 3.4.4's closed-loop optimization claims checkable. Per-task streams, the walk-through, and both result tables are in Appendix H of the supplementary material; the headline: every task's guard survives compilation (never dropped outright, the reduction being fusion), the first realization pays all model calls and every subsequent one hits the cache (91.7%, 11 of 12 realizations by construction), hot-path adoption lands at repeat 7 executing the shrunken six-instruction stream, and all reports come out byte-identical (12/12 plus fresh-stack SHA-256 equality); the compile shrink is the only shape-dependent quantity (12.5–25.0%)—the composition behavior is not an artifact of one stream's structure.

The triage task's instruction stream (seven instructions over sandbox-resident files—side-effectful first and last, the planted dead guard `check_syntax`, the cacheable LLM-type `classify_signature` behind a deterministic stand-in, pure normalization/aggregation between) is representative; the full per-task streams are in Appendix H of the supplementary material.

Calibrating the synthetic headline. Section 5.4 reports 65.8% trace shrinkage on synthetic traces with planted dead instructions; the four realistic tasks show **12.5–25.0%** (one planted guard per stream, plus one additional live-pure fusion pair in csv-etl-anomaly). We state both figures rather than privilege either: 12.5–25.0% reflects the planted dead-check density (one guard per seven- to eight-instruction stream)—a stated density assumption, not observed ordinary-pipeline waste—while the synthetic figure is an upper bound under the skew/planting parameters we control, not a typical-case estimate. Constructing a realistic (non-synthetic) dependency shape also served the audit purpose, exposing two grouping/fusion soundness gaps—side-effectful instructions opening dispatch groups, fused instructions losing their external dependencies—now fixed with regression tests; Section 5.3's dispatch-folding numbers reflect the corrected (more conservative) grouping.

Structural invariance. The headline repeats structurally in all four tasks: the eliminated set is empty everywhere (the reduction is fusion, with a second adjacent pure pair also fusing in csv-etl-anomaly—hence its 25.0% shrink against a single planted guard), and the 91.7% hit rate is shared structurally (one cacheable step, identical seeded draws). Reuse and closed-loop adoption are thus invariant across the four dependency shapes under the fixed protocol; only the guard/fusion shrink varies—exactly the evidence the single-task demonstration could not provide.

5.9 Limitations and Threats to Validity

We enumerate the limitations of the evaluation explicitly:

1. **Mock backend, not a live LLM.** The mock charges a constant 80 ms per simulated call and returns deterministic responses; beyond counting, it covers Section 5.4's replay verification (final-state identity, 5/5 seeds) and Section 5.2's naive-cache baseline (same instrumentation). The real-gateway experiments extend coverage to the cache's hit/miss reproduction with measured time/token savings (Section 5.7) and the compiled, hot-path-adopted stream's zero new gateway calls and byte-identical artifacts (Section 5.7.1). Still not covered: multi-seed behavior beyond the two-seed reduced-scale rerun (Section 5.7), real deployment, caching under model nondeterminism *across prompt variations*, streaming responses, provider rate limits, and live-model task quality beyond the Section 5.7.1 rubric diagnostics, now run on both a reasoning model and two completing models (the four-task classification steps remain deterministic rule-based stand-ins).

2. **Synthetic workloads.** Zipf-distributed synthetic tasks are a reasonable model of skewed agent workloads, but a model; Sections 5.8 and 5.10 only partially address this. The cache-reduction figures (59.5–79.6%) are exact for these workload families and should be read as *skew-sensitivity curves*, not universal savings; uniformly novel tasks would see near-zero benefit. Section 5.8's four-shape characterization calibrates the 65.8% synthetic headline downward with its 12.5–25.0% per-task compile shrink and shows uniform reuse and closed-loop adoption under the fixed protocol—but the four tasks remain seed-controlled, hand-constructed streams, not a production workload study.

3. **No production end-to-end evaluation.** We have not deployed AgentISA in a production setting; the paper therefore contains no latency-under-load, throughput, availability, or dollar-cost measurements, and makes no such claims.

4. **No real cost data (token usage measured; pricing MODELED on measured tokens).** All latency figures in Sections 5.2–5.5 are simulated (avoided calls × 80 ms); Section 5.7 measures real wall-clock time and gateway-reported tokens at reduced scale, and Section 5.7.1 prices its own measured tokens under three modes—no optimization, AgentISA result caching, and a modeled provider prompt cache with a parameterized cache-read discount (default 0.1). The costs are MODELED-on-measured: tokens are measurements, prices are configurable list-price stand-ins (the gateway's actual pricing is unknown), and no dollar-savings claim is made. Section 5.2's baseline is an *exact-match* response cache (parity reported); head-to-head against *approximate* (semantic/embedding) caching (GPTCache's approximate mode—explicitly out of scope) and a priced deployment study under real pricing remain future work.

5. **Program-size metrics, not wall-clock time.** Effective-instruction and dispatch-count reduction are structural metrics: they correlate with—but do not equal—wall-clock improvements, which depend on per-instruction cost, available parallelism, and dispatch overhead.

6. **Descriptive statistics only.** Synthetic configurations use five seeds (Section 5.1.3); the real-gateway and system-level runs are single-seed by design (Section 5.7's two-seed rerun excepted); no hypothesis tests, confidence intervals, or significance claims are made—standard deviations indicate stability across seeds, nothing more.

7. **Functional, not performance, validation of coordination.** The 43 multi-agent tests verify behavioral correctness under deterministic conditions, not throughput or scalability.

Despite these limitations, the evaluation demonstrates, under fully disclosed and reproducible conditions, that each optimization mechanism is implemented, active, and produces the claimed structural effects on workloads designed to expose them.

5.10 Real-Trace Workload Characterization

The synthetic-workload threat of Section 5.9 motivates an independent census: are Zipf structure and reuse artifacts of the hand-built benchmarks? We analyze 100 task traces from a seeded random-task stress battery—randomly shaped instruction streams executed through the production engine, not hand-picked—totaling 792 steps and 12 instruction signatures (protocol and tables: Appendix I of the supplementary material). Frequency ranks fit log-rank/log-frequency OLS with $s = 0.593$ ($R^2 = 0.971$): decay flatter than classical Zipf yet strongly log-linear. Corpus-level repeated instances reach 98.5% (780/792) while per-workflow repetition averages 18.3%—reuse is cross-task vocabulary, not intra-task duplication. Shape census: 79 fan-out, 21 linear, 0 fan-in, 0 branch (7.9 steps/workflow); 152 dead-end steps exactly equal the planted guards; 106 of 692 adjacent pairs (15.3%) pass the compiler's fusibility gate, all CONTROL_FLOW. Disclosure: the corpus is generator-produced, not production logs; the census characterizes workload structure only.

5.11 Mechanism Extensions

Six experiments exercise opt-in mechanisms (Appendix I). **Capability calibration:** extracted domain scores correlate with mock-backend behavior on declaration-injection pools (Pearson r): memory +0.948 (hit rate), coordination −0.977 (dispatch compression), reasoning −0.965 (fusion reduction), generation +0.790; optimization/security ± 1.000 (declared-and-enforced); 2/8 domains, 9/17 metrics N/A (probe-evidenced). **DP tree-pattern selection:** covering cost ≤ heuristic on 20/20 random IR trees (ratio 0.361; time 0.950); gap size is synthetic-pricing-dependent. **EMA hot-path detection:** 3/3 rotated variants promoted (exact: 0/3; Zipf recall 0.867 ± 0.298), zero false promotions on one-shot diversity. **Strict security:** of 1,000 paired tasks, fail-closed denies 400 policy-gap tasks fail-open admits; 0/200 normal-task false denials. **Cache namespacing:** 100% invalidation under template/decoding drift (legacy keys 100% stale); result cache ≤ prompt cache (modeled) in 195/195 token-mix cells. **Parameterized replay:** all 1,050 unseen-parameter replays match the oracle exactly. Chain-enriched workloads amplify BASIC→ADVANCED dispatch reduction to 79.1% (random-slots control 46.7%), AGGRESSIVE differing only in the advisory memory plan.

6 Discussion

6.1 System Limitations and Challenges

1. **Instruction-set coverage**: complex reasoning patterns are not yet naturally expressible (domain-specific extensions are likely needed), and maintaining consistency of pre/post-conditions and typing rules across extensions remains an open engineering challenge.

2. **Evaluation scope**: the quantitative evidence is primarily microbenchmark-based (Sections 5.2–5.5), supplemented by real-gateway spot checks (Sections 5.7–5.7.1) and the four-task characterization (Section 5.8); production end-to-end evaluation remains required before any deployment claim (Section 5.9).

3. **Performance bottlenecks**: cache memory consumption under large instruction universes; synchronization overhead in barrier-coordinated multi-agent execution; recompilation cost when hot-path thresholds are misconfigured; the *default* hot-path detector keys on exact sequence identity, so trace diversity never accumulates counts (the opt-in EMA detector of Section 4.2.2 is the shipped remedy, still exact-path-keyed, so its denominator can be floored by tails of one-shot paths unless the tracked-path cap evicts them); hot-path state is bounded by the shipped LRU cap on tracked paths (default 1,000), though recency-only eviction can drop a *hot* path's state, not just cold ones.

4. **Integration challenges**: adapter maintenance as provider function-calling surfaces evolve; version management of learned instructions across LLM updates.

6.2 Security and Privacy Considerations

AgentISA's implemented security layer is the `SecurityMonitor` of Section 5.6, consulted by the coordinator before every assignment, with violations timestamped. One shipped default deserves a call-out: **resource access is open by default** and restrictable by configuration—a fail-open stance shifting the burden onto the policy writer. The hardened inversion ships as an opt-in strict mode (`SecurityMonitor(..., strict=True)`, propagated by `AgentCoordinator(..., security_strict=True)`; default `False`, fail-open checks unchanged): strict denies unregistered resources, malformed resource-access policies, and unknown data sensitivity, recording each in the timestamped violation history with a `(strict mode)` marker (`tests/core/isa/test_security_failclosed.py`). A paired posture matrix (1,000 tasks per mode, five seeds, fresh `SecurityMonitor` and `AgentCoordinator` per task; Section 5.11) quantifies the difference on the real code paths: fail-open admits all 400 unknown-sensitivity and incomplete-policy tasks that strict denies; both deny the 200 unregistered-resource tasks, strict adding a distinguishable reason class; strict adds zero

false denials on the 200 well-specified tasks; every strict denial is recorded. The 43 functional tests validate the implemented checks under configured policies, not the default posture; nothing in this paper constitutes a security certification. Beyond what is implemented, hardening considerations—not claims about the current system: data privacy (end-to-end data-flow tracking, privacy-preserving computation; Section 6.3.6), secure multi-agent communication channels, audit-log integrity, and externally audited compliance controls all remain future work; the system provides enforcement hooks, not certification.

6.3 Future Research Directions

6.3.1 Theoretical Foundations.

1. **Formal semantics of agent instructions**: consolidating Section 3.1 and Appendix B into one standalone semantics—instruction language, state space, reduction rules, type system; nothing in it is new machinery.
2. **Compositional reasoning**: a parallel-composition rule whose disjointness side condition mirrors the $\|$-typing rule of Definition 3.6, enabling modular verification and temporal property preservation (rule in Appendix E.1 of the supplementary material).
3. **Type theory for agent capabilities**: a capability type discipline with Basic/Composite/Emergent kinds and a composition rule $a \circ b : \text{cap}_1 \otimes \text{cap}_2$ (Appendix E.1).

6.3.2 Advanced Instruction Set Evolution.

1. **Self-modifying instructions**—a formal model for instruction evolution, $I_{t+1} = f(I_t, \text{Performance}_t, \text{Context}_t)$ (f is left abstract: the sketch names the inputs, not a concrete update rule), under semantic-preservation, performance-monotonicity, and safety constraints.
2. **Context-aware optimization**—for each observed context, predict the instruction set's performance against a threshold, and if it falls short, select and accumulate a best rewrite variant. The programmatic sketch (OPTIMIZE-CONTEXT) is listed in Appendix D.9; its primitives are future work, not implemented machinery.
3. **Cross-model instruction mapping**: a universal mapping framework $\phi_{A \to B} : I_A \to I_B$ with semantic preservation, performance bounds, and capability awareness stated as goals, not delivered guarantees.

6.3.3 Theoretical Performance Bounds.

1. **Asymptotic analysis**: deriving cost bounds per composition operator—when does iterative composition i_1^* admit a closed-form cost model, and under what reuse structure does caching yield amortized $O(1)$ effective cost per task? The microbenchmarks (Section 5) measure the constants empirically; the asymptotic theory remains open.
2. **Resource consumption models**: a calibrated model $R(i) = \alpha C_{cpu}(i) + \beta C_{mem}(i) + \gamma C_{io}(i)$ with platform-specific coefficients fitted against real deployments rather than assumed (Appendix E.1).

6.3.4 Capability Assessment: Extensions. The instruction-level capability assessment subsystem (`CapabilityProfile`, Section 3.5) now ships at the static layer; its wiring gap remains disclosed there. Two extensions remain future work: (i) **benchmark-calibrated metrics**—a controlled first step now ships (Section 5.11), but correlating the scores with *end-to-end* benchmark performance, in the manner of function-calling evaluations [17] and agent benchmarks [29], and grounding the weights w_k in such measurements rather than configuring them by hand, remain open; and (ii) **cross-model comparison matrices**, extending the per-agent profile to a pairwise framework $\Delta(M_1, M_2)$ over capability dimensions, supporting systematic model selection for a given instruction workload.

6.3.5 Emergent Behavior Analysis.

1. **Collective intelligence modeling**: the open question is a formal metric for when multi-agent composition exceeds the sum of its members' capabilities—emergence made measurable, rather than an appeal to an undefined φ over unioned capability sets.
2. **Learning transfer mechanisms**: knowledge transfer across agents composes the mining-extraction-validation pipeline of Section 3.3 rather than introducing new machinery—patterns mined from a source agent's traces would be adapted and revalidated before promotion; the validation obligation transfers unchanged.

6.3.6 Security and Privacy Extensions.

1. **Information flow control**: the classical noninterference property $\text{Secure}(P) \iff \forall s_1, s_2 : L(s_1) = L(s_2) \implies L([\![P]\!]s_1) = L([\![P]\!]s_2)$, where L assigns security levels to states and outputs, stated for terminating runs (divergence and channel leakage out of scope; Appendix E.1).
2. **Privacy-preserving computation**: a direction, not a delivered mechanism—if agent computations over sensitive data were routed through per-step differentially private mechanisms, the overall loss would follow a composition theorem over the per-mechanism (ϵ_i, δ_i) parameters. We deliberately state no bound: which composition theorem applies, and which concrete mechanisms the runtime would expose, are open design choices; the instruction contract is the natural place to carry such annotations once the mechanisms exist.

None of the frameworks above is implemented today; Sections 6.3.3 and 6.3.4 are stated precisely enough to be falsifiable, while the remaining sketches are programmatic—guidance from controlled-setting-validated infrastructure toward a deployable, formally verified agent platform.

7 Conclusion

This paper presented AgentISA, an instruction set architecture that treats LLM agent behavior as a first-class compilable artifact. Our contributions are:

1. A formal instruction set with compositional semantics, a type system, and pre/post-conditions, with sanity results as design rationale (Theorems 3.1–3.3) and semantic-preservation arguments for the optimization and scheduling transformations (Theorems 3.4–3.6).
2. Agentic Calling, which lifts provider-specific function calling into reusable, provider-independent, cacheable agent instructions.
3. Trace-based instruction learning—frequency mining with deterministic action-signature grouping, developer-controlled promotion, and composite replay, with shipped schema-checked parameter binding (unseen-parameter replays exact on 1,050/1,050 cases; constructor-baked parameters excepted; Section 5.11).
4. A compiler and runtime applying classical optimizations—fusion, dead-code elimination, scheduling, and dynamic hot-path specialization—to agent instruction streams.
5. A multi-agent coordination layer with capability-based selection and a security monitor (authentication, authorization, data-sensitivity policies), now with an opt-in fail-closed mode.

The evaluation shows the optimizations are real and measurable under controlled conditions: 59.5–79.6% simulated-call reduction from caching, 34.2% instruction-count reduction from static compilation, 65.8% trace shrinkage from hot-path optimization (Sections 5.2–5.5), real-gateway validations of the cache and the compiled pipeline (Sections 5.7–5.7.1; integration and cache-mediated-identity layers only), a four-task system-level characterization (Section 5.8), a 100-trace workload census (Section 5.10), six controlled mechanism experiments (Section 5.11), and 43 functional tests for the multi-agent layer. These are microbenchmark and controlled-setting results, not production end-to-end measurements; the threats to validity of Section 5.9 stand.

Implementation status of the contributions. For auditability, the five contributions above split by status: *implemented and exercised by the benchmarks*—the instruction set and contracts, Agentic Calling, result caching, fusion/DCE/grouping, hot-path detection and adoption (exact default), the coordination layer with injected-score capability selection, and the security monitor (fail-open default; 43 tests); *implemented with known limits, exercised by the Section 5.11 mechanism experiments but not the Section 5.2–5.5 benchmark arms*—the opt-in DP tree-pattern selector (default heuristic), the EMA hot-path detector (LRU-bounded tracked-path state; default exact), the strict cache namespace (default params-only key), the strict security mode, parameter-binding composite replay (mined via the learner's threshold-validated path, not the approval gate; constructor-baked parameters not rebindable), and capability metric extraction (static, declared-metadata only; coordinator wiring still injected); *implemented with known limits, still unexercised by any Section 5 experiment*—the bounded checker (pure instructions, bounded state spaces only) and runtime context validation (deletion blind spot disclosed); *specified, not implemented*—cycle-accurate pipeline scheduling, block-granular trace formation, adaptive performance-triggered recompilation, and symbolic-equivalence checking. Theorems 3.4–3.6 are conditional arguments for the implemented passes; the specified components carry no verification claim.

AgentISA's longer-term significance is architectural: an instruction-level contract for agent behavior makes agent systems amenable to the discipline that transformed hardware and compiler engineering—precise specification, portability across backends, and semantics-preserving optimization. Establishing the corresponding gains on production workloads, and mechanically verified implementations of the conditional sanity and semantic-preservation arguments, is the immediate next step.

Artifact Availability

Full proofs of Theorems 3.1–3.2 and 3.4–3.6, the component listings, the future-direction sketches, the artifact-regeneration command lines, the priced-comparison protocol, the four-task characterization detail, and the extension-experiment protocols and detailed tables behind Sections 5.10–5.11 (supplementary Appendices A–I, built from this repository as `docs/AgentISA_supplementary.pdf`) are in the supplementary material; the commands are mirrored in the repository's top-level `REPRODUCING.md`. All artifacts are in the repository accompanying this submission (anonymized for review), and every number in Section 5 is generated from the exact repository state whose build produces this PDF (`REPRODUCING.md` records the submission-time `git rev-parse HEAD`, the dependency lock file, and the Python/system versions). A fresh clone reproduces the full-suite count with `python -m pytest tests/ -q` (1,333 passed, 14 skipped at run time), the Section 5.6 functional-test counts, and, via the Appendix F command lines, every results table from scratch—real-gateway runs included, with a mock-backend fallback for environments without gateway credentials. `REPRODUCING.md` also maps each paper claim to the implementing module, the test that exercises it, and the raw-results file that backs its numbers, flagging specified-only components; the artifact is inventoried per file in Appendix F.7.

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